

# CM0246 Discrete Structures

## Mathematical Induction

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# Preliminaries

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## Textbook

Kenneth H. Rosen ([1988] 2004). Matemática Discreta y sus Aplicaciones.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Motivation

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## Exercise

Conjecture a formula for the sum of the first  $n$  positive odd integers.

# Motivation

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## Exercise

Conjecture a formula for the sum of the first  $n$  positive odd integers.

## Question

Let  $P(n)$  be a propositional function. How can we proof that  $P(n)$  is true for all  $n \in \mathbb{Z}^+$ ?

# Principle of Mathematical Induction

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## Proof by mathematical induction

Let  $P(n)$  be a propositional function.

To prove that  $P(n)$  is true for all  $n \in \mathbb{Z}^+$ , we must make two proofs:

- **Basis step:** Prove  $P(1)$

# Principle of Mathematical Induction

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To prove that  $P(n)$  is true for all  $n \in \mathbb{Z}^+$ , we must make two proofs:

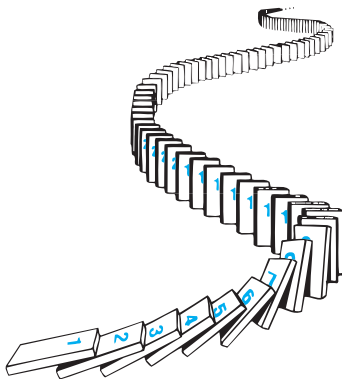
- **Basis step:** Prove  $P(1)$
- **Inductive step:** Prove  $P(k) \rightarrow P(k + 1)$  for all  $k \in \mathbb{Z}^+$

$P(k)$  is called the **inductive hypothesis**.

# Principle of Mathematical Induction

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How mathematical induction works<sup>†</sup>



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<sup>†</sup>Figure source: (Rosen [1988] 2012, § 5.1, Fig. 2).

# Principle of Mathematical Induction

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Definition (principle of mathematical induction)

Inference rule version:

$$\frac{\begin{array}{c} [P(k)] \\ \vdots \\ P(1) \quad P(k+1) \end{array}}{P(n)} \text{ (PMI)}$$



# Principle of Mathematical Induction

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Definition (principle of mathematical induction)

Inference rule version:

$$\frac{\begin{array}{c} [P(k)] \\ \vdots \\ P(1) \quad P(k+1) \\ \hline P(n) \end{array}}{\text{(PMI)}}$$

Axiom (or theorem) version: Let  $P$  be a propositional function (predicate). Then

$$[P(1) \wedge \forall k(P(k) \rightarrow P(k+1))] \rightarrow \forall n P(n) \quad \text{(PMI)}$$

# Principle of Mathematical Induction

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Methodology for proving by mathematical induction

# Principle of Mathematical Induction

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1. State the propositional function  $P(n)$ .
2. Prove the basis step, i.e.  $P(1)$ .
3. Prove the induction step, i.e.  $\forall k(P(k) \rightarrow P(k+1))$ .

**Remark:** In this proof you need to use the inductive hypothesis  $P(n)$ .

# Principle of Mathematical Induction

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## Methodology for proving by mathematical induction

1. State the propositional function  $P(n)$ .
2. Prove the basis step, i.e.  $P(1)$ .
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**Remark:** In this proof you need to use the inductive hypothesis  $P(n)$ .

4. Conclude  $\forall n P(n)$  by the principle of mathematical induction.

# Principle of Mathematical Induction

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## Example

Prove that the sum of the first  $n$  odd positive integers is  $n^2$ .<sup>†</sup>

Whiteboard.

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<sup>†</sup>Historical remark. From 1575, it could be the first property proved using the PMI (Gunderson 2011, § 1.8).

# Principle of Mathematical Induction

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## Example

Prove that if  $n \in \mathbb{Z}^+$ , then

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$



# Principle of Mathematical Induction

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## Proof

1.  $P(n)$ :  $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$

# Principle of Mathematical Induction

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## Example

Prove that if  $n \in \mathbb{Z}^+$ , then

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## Proof

1.  $P(n)$ :  $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ .
2. Basis step  $P(1)$ :  $1 = \frac{1(1+1)}{2}$ .

(continued on next slide)

# Principle of Mathematical Induction

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## Proof (continuation)

### 3. Inductive step:

Inductive hypothesis  $P(k)$ :  $1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$ .

Let's prove  $P(k+1)$ :

$$\begin{aligned} 1 + 2 + 3 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) && \text{(by IH)} \\ &= (k+1) \left( \frac{k}{2} + 1 \right) && \text{(by arithmetic)} \\ &= \frac{(k+1)(k+2)}{2} && \text{(by arithmetic)} \end{aligned}$$

# Principle of Mathematical Induction

## Proof (continuation)

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### 4. $\forall n P(n)$ by the principle of induction mathematical. ■

# Principle of Mathematical Induction

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## Example

Prove that if  $n \in \mathbb{N}$ , then

$$2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

(continued on next slide)

# Principle of Mathematical Induction

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## Proof

1.  $P(n): 2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$

# Principle of Mathematical Induction

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## Proof

1.  $P(n)$ :  $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$
2. Basis step  $P(0)$ :  $2^0 = 1 = 2^{0+1} - 1$ .

# Principle of Mathematical Induction

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## Proof

1.  $P(n)$ :  $2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$

2. Basis step  $P(0)$ :  $2^0 = 1 = 2^{0+1} - 1$ .

3. Inductive step:

Inductive hypothesis  $P(k)$ :  $2^0 + 2^1 + 2^2 + \cdots + 2^k = 2^{k+1} - 1$

Let's prove  $P(k+1)$ :

$$\begin{aligned} 2^0 + 2^1 + 2^2 + \cdots + 2^k + 2^{k+1} &= 2^{k+1} - 1 + 2^{k+1} && \text{(by IH)} \\ &= 2(2^{k+1}) - 1 && \text{(by arithmetic)} \\ &= 2^{k+2} - 1 && \text{(by arithmetic)} \end{aligned}$$



# Principle of Mathematical Induction

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## Proof

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Inductive hypothesis  $P(k)$ :  $2^0 + 2^1 + 2^2 + \cdots + 2^k = 2^{k+1} - 1$

Let's prove  $P(k+1)$ :

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4.  $\forall n P(n)$  by the principle of induction mathematical. ■

# Strong Induction

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Proof by strong (or course-of-values) induction

Let  $P(n)$  be a propositional function.

To prove that  $P(n)$  is true for all  $n \in \mathbb{Z}^+$ , we must make two proofs:

- **Basis step:** Prove  $P(1)$

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The (strong) inductive hypothesis is given by

$$P(j) \text{ is true for } j = 1, 2, \dots, k.$$

# Strong Induction

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## Definition ([strong induction])

Inference rule version:

$$\frac{P(1) \quad \forall k[(P(1) \wedge P(2) \wedge \cdots \wedge P(k)) \rightarrow P(k+1)]}{\forall n P(n)} \text{ (strong induction)}$$

# Strong Induction

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Example (a part of the fundamental theorem of arithmetic)

Prove that if  $n$  is an integer greater than 1, either is prime itself or is the product of prime numbers.

(continued on next slide)

# Strong Induction

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## Proof

1.  $P(n)$ :  $n$  is prime itself or it is the product of prime numbers.

# Strong Induction

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## Proof

1.  $P(n)$ :  $n$  is prime itself or it is the product of prime numbers.
2. Basis step  $P(2)$ : 2 is a prime number.



# Strong Induction

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Inductive hypothesis:  $P(j)$  is true for  $j = 1, 2, \dots, k$ .

# Strong Induction

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2. Basis step  $P(2)$ : 2 is a prime number.
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Inductive hypothesis:  $P(j)$  is true for  $j = 1, 2, \dots, k$ .  
Let's prove that  $k + 1$  satisfies the property:

# Strong Induction

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## Proof

1.  $P(n)$ :  $n$  is prime itself or it is the product of prime numbers.
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Inductive hypothesis:  $P(j)$  is true for  $j = 1, 2, \dots, k$ .  
Let's prove that  $k + 1$  satisfies the property:  
3.1 If  $k + 1$  is a prime number then it satisfies the property.

# Strong Induction

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## Proof

1.  $P(n)$ :  $n$  is prime itself or it is the product of prime numbers.
2. Basis step  $P(2)$ : 2 is a prime number.
3. Inductive step:

Inductive hypothesis:  $P(j)$  is true for  $j = 1, 2, \dots, k$ .

Let's prove that  $k + 1$  satisfies the property:

3.1 If  $k + 1$  is a prime number then it satisfies the property.

3.2 If  $k + 1$  is a composite number:

$k + 1 = ab$  where  $2 \leq a \leq b < k + 1$ . Since  $P(a)$  and  $P(b)$  by the inductive hypothesis, then  $P(k + 1)$ .

# Strong Induction

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## Proof

1.  $P(n)$ :  $n$  is prime itself or it is the product of prime numbers.

2. Basis step  $P(2)$ : 2 is a prime number.

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Inductive hypothesis:  $P(j)$  is true for  $j = 1, 2, \dots, k$ .

Let's prove that  $k + 1$  satisfies the property:

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$k + 1 = ab$  where  $2 \leq a \leq b < k + 1$ . Since  $P(a)$  and  $P(b)$  by the inductive hypothesis, then  $P(k + 1)$ .

4.  $P(n)$  is true for all integer  $n$  greater than 1 by strong induction. ■

# First-Order Peano Arithmetic

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## Axioms of first-order Peano arithmetic<sup>†</sup>



Giuseppe Peano  
(1858 – 1932)

$$\forall n. 0 \neq n'$$

$$\forall m \forall n. m' = n' \rightarrow m = n$$

$$\forall n. 0 + n = n$$

$$\forall m \forall n. m' + n = (m + n)'$$

$$\forall n. 0 * n = 0$$

$$\forall m \forall n. m' * n = n + (m * n)$$

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<sup>†</sup>See, for example, (Hájek and Pudlák [1993] 1998).

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$$\forall n. 0 * n = 0$$

$$\forall m \forall n. m' * n = n + (m * n)$$

For all formulae  $A$ ,

$$[A(0) \wedge (\forall n. A(n) \rightarrow A(n'))] \rightarrow \forall n A(n)$$

<sup>†</sup>See, for example, (Hájek and Pudlák [1993] 1998).

# First-Order Peano Arithmetic

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## Theorem

The principle of mathematical induction and strong induction are equivalent.<sup>†</sup>

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<sup>†</sup>See, for example, (Gunderson 2011).



# References

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- David S. Gundersen (2011). Handbook of Mathematical Induction. Chapman & Hall (cit. on pp. 15, 40).
- Petr Hájek and Pavel Pudlák [1993] (1998). Metamathematics of First-Order Arithmetic. Second printing. Springer (cit. on pp. 38, 39).
- Kenneth H. Rosen [1988] (2004). Matemática Discreta y sus Aplicaciones. Trans. by José Manuel Pérez Morales, Julio Moro Carreño, Ana Isabel Lías Quintero and Pedro Antonio Ramos Alonc. 5th ed. McGraw-Hill (cit. on p. 2).
- [1988] (2012). Discrete Mathematics and Its Applications. 7th ed. McGraw-Hill (cit. on p. 7).